CURRICULUM VITAE

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PROFILE

M.Sc. student in Artificial Intelligence for Industrial Applications (OTH Amberg-Weiden) with a B.Sc. in Mechanical Engineering and industry experience in aviation design and production automation. Focus on computer vision, LLMs/NLP and AI for industrial processes. I build models that hold up outside the notebook: carefully evaluated, explainable and small enough to deploy.

WORK EXPERIENCE

10/2026 – present

Working Student (Werkstudent)

syskomp gehmeyr GmbH · Amberg, Germany · part-time

- Apply AI methods to engineering and industrial workflows in a mechanical engineering environment.
- Build and evaluate Python prototypes with an emphasis on traceable, validated results.
- Work with engineering colleagues to turn requirements into practical AI tooling.

08/2023 – 09/2024

Design Engineer

KNSI Aviation Engineering · Negombo, Sri Lanka · full-time

- Managed compliance data workflows and technical documentation with SQL and Excel to aviation standards.
- Built automated SQL/Excel reporting templates that cut weekly reporting time by 30 %.
- Analysed production data to optimise workflows, raising process efficiency by 40 %.
- Led the digitalisation of legacy documentation for faster cross-department data access.

02/2022 – 11/2022

Internship Trainee, Mechatronics & Data

CBL Group (Ceylon Biscuits Limited) · Pannipitiya, Sri Lanka · internship

- Contributed to process optimisation through data-driven solutions and automation projects.
- Supported end-of-line automation with articulated and parallel robots.
- Designed and prototyped 3D-printed automation components, reducing external sourcing costs and lead times.

EDUCATION

03/2025 – present

M.Sc. Artificial Intelligence for Industrial Applications

OTH Amberg-Weiden · Amberg, Germany · graduation expected 2027

- Modules: Computer Vision, Deep Vision, Deep Learning, NLP & Information Retrieval, Autonomous Robotics, AI Security & Privacy, Machine Learning.

10/2024 – 02/2025

Preparation for Master's studies in Germany

Visa process, relocation to Germany, German language course

11/2019 – 05/2023

B.Sc. (Hons) Mechanical Engineering, Minor in Mechatronics

University of Sri Jayewardenepura · Sri Lanka · GPA 3.27 / 4.0

- Thesis: CNN surrogate model that predicts the aerodynamic drag of 2D shapes from images (replaces CFD runs).
- Project: design and build of a CoreXY 3D printer integrating mechanical design and machine learning.

SELECTED PROJECTS

2026 **Removing Jailbreak Backdoors from LLMs** GitHub ↗
Removed a known backdoor from Qwen2.5-1.5B with four-group hybrid supervised fine-tuning (LoRA) on one 8 GB GPU: attack success rate 95 % → 1.7 % at 84 % benign accuracy; probes show the trigger is hidden, not erased. [PyTorch · Hugging Face · PEFT/LoRA]

2026 **Pneumonia Detection: CNN vs ViT vs Hybrid** GitHub ↗
Compared ResNet-50, ViT-Base and TinyViT under identical conditions (5-fold CV ensembles): TinyViT reached 94.9 % test accuracy (F1 0.948); assessed how faithful the Grad-CAM explanations are.
[PyTorch · timm · Grad-CAM]

2026 **X-ray Casting Defect Detection** GitHub ↗
Fine-tuned YOLOv8n on GDXray castings with a series-level split: test mAP50 0.545, precision 0.748; failure analysis by defect size; ONNX export running at 60 ms per image on CPU. [YOLOv8 · OpenCV · ONNX]

2025/26 **QuantumCircuit Dataset Builder** GitHub ↗
Automated arXiv mining pipeline with a fine-tuned SciBERT caption classifier (F1 0.954) that produced 250 metadata records and 160 extracted circuit images for image-to-text training.
[Transformers · BeautifulSoup · PyMuPDF]

2026 **LayoutLMv3 Form Understanding** GitHub ↗
Fine-tuned LayoutLMv3 for key-value extraction from scanned forms (FUNSD): entity-level F1 0.896, with a Tesseract OCR path for unannotated documents. [LayoutLMv3 · Hugging Face · Tesseract]

2025/26 **Stacking Ensemble & Edge-Ready Rule Model** GitHub ↗
Optuna-tuned XGBoost/LightGBM/CatBoost stack reaching R^2 0.848 on 273 features; reverse-engineered three if-else rules (R^2 0.915) that run on an edge device without ML libraries. [scikit-learn · XGBoost · Optuna]

2025/26 **Autonomous Robotics & Digital Twin** GitHub ↗
ROSbot that explores unknown mazes with LiDAR/depth SLAM, frontier exploration, A* and PID (4 of 5 mazes completed); hospital delivery digital twin with TurtleBot3, MQTT and a Flask control centre.
[Webots · SLAM · MQTT]

2023 **CNN Surrogate for Aerodynamic Drag (B.Sc. thesis)** GitHub ↗
CNN that predicts the drag coefficient of 2D shapes from images, trained on 12,000 simulated shapes: 0.06–2.77 % relative error on real shapes including NACA airfoils. [TensorFlow/Keras · CFD]

TECHNICAL SKILLS

Programming	Python (advanced), C++, SQL, Git/GitHub, REST APIs, pytest, Linux/Bash
ML & Deep Learning	PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna
Computer Vision	OpenCV, YOLOv8, U-Net, DeepLabV3+, Vision Transformers, Grad-CAM, ONNX
NLP & Generative AI	Hugging Face Transformers, LLM fine-tuning (LoRA/PEFT), RAG, LangChain, SciBERT, LayoutLMv3
Robotics & Engineering	Webots, SLAM, path planning (A*), PID control, MQTT, digital twins, CAD/CAM, CFD
Data & Tools	pandas, NumPy, MongoDB, Power BI, Flask, Jupyter, Weights & Biases, LaTeX

LANGUAGES

English	C1 · professional working level
German	B1 · actively improving
Sinhala	Native language

CERTIFICATIONS & TRAINING

09/2025 · Bavarian–Czech AI Summer School: "Controlling Robots with LLMs" (32 h) · Speinshart, Germany

Deep Learning Specialization · DeepLearning.AI (Coursera)

Natural Language Processing Specialization (4 courses) · DeepLearning.AI

AI for Medicine Specialization (3 courses) · DeepLearning.AI

TALKS, ACTIVITIES & INTERESTS

Talk	Adaptive-RAG (NAACL 2024): 25-minute scientific talk, AI Conference module, OTH Amberg-Weiden, summer 2026.
Entrepreneurship	VYROBA TECHS: designed and built a CoreXY 3D printer and ran a small custom-parts business during the pandemic.
Football	Player at FC Amberg; former captain of the faculty and university teams; University Colours Award 2022.